



Data Analytics Capstone

Machine Learning-Based Forecasting of Urban Air Quality Using Environmental and Meteorological Data

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Executive Summary

01



OBJECTIVE & SCOPE

Compare 8 supervised ML regression models for one-hour-ahead PM2.5 forecasting.

02



IMPORTANCE

Support air-quality monitoring, early warning, and environmental decision-making.

03



DATA & TECHNOLOGY

409,158 forecasting observations from 12 Beijing monitoring stations, March 2013–February 2017.
Python-based workflow using Scikit-learn, LightGBM, XGBoost, CatBoost, and SHAP.

04



FINAL RESULT

LightGBM achieved the best test performance: MAE 9.67 $\mu\text{g}/\text{m}^3$ | RMSE 18.26 $\mu\text{g}/\text{m}^3$ | R^2 0.9522.

05



INTERPRETABILITY & USAGE

Current-hour PM2.5 was the dominant predictor (98.17% gain-based importance), supporting a proposed one-hour-ahead forecasting and decision-support framework.

Gap Analysis

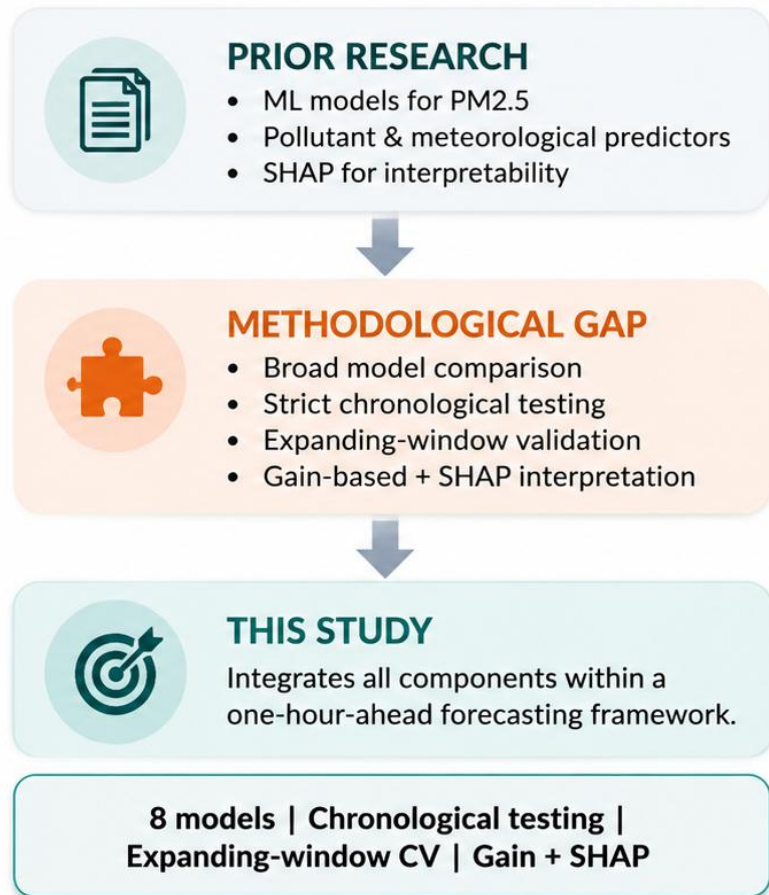
PRIOR RESEARCH ESTABLISHES

- Machine learning can capture nonlinear PM2.5 relationships, with ensemble and boosting models showing strong predictive performance.
- Pollutant, meteorological, temporal, and location variables have been used as predictors; SHAP can improve model interpretability.

METHODOLOGICAL GAP

Comparatively fewer studies integrate:

- Broad model comparison
- Strict chronological testing
- Expanding-window validation
- Gain-based + SHAP interpretation within a one-hour-ahead forecasting framework.



Literature Matrix & Research Question Linkage

Key studies that informed the problem, features, model selection, and interpretability in this study.

#	Author(s) & Year	Key Finding / Contribution	Influence on This Study	Linked RQ(s)
1	Breiman (2001)	Random Forest reduces overfitting and handles non-linear relationships.	Informed inclusion of Random Forest as a candidate model.	RQ3 RQ4
2	Friedman (2001)	Gradient Boosting is an effective ensemble technique for regression.	Informed inclusion of Gradient Boosting (e.g., XGBoost) as a candidate model.	RQ3 RQ4
3	Chen & Guestrin (2016)	XGBoost is a scalable and efficient gradient boosting algorithm.	Supported use of XGBoost for model development and comparison.	RQ3 RQ4
4	Lundberg & Lee (2017)	SHAP provides consistent and interpretable feature importance.	Used SHAP for model interpretability and to identify key predictor variables.	RQ4
5	Zhang et al. (2017)	Assessed Beijing's PM2.5 pollution and its relationship with meteorological factors.	Supported inclusion of current-hour pollutant and meteorological variables (e.g., wind direction, weather).	RQ1
6	Hastie et al. (2021)	Comprehensive guide to statistical learning and model evaluation.	Informed model selection, evaluation approach, and statistical significance testing.	RQ3 RQ4
7	Begum et al. (2018)	Time-series cross-validation methods improve forecasting reliability.	Used time-series validation to evaluate forecasting performance.	RQ2
8	Duan et al. (2023)	ML models with meteorological and pollutant variables can effectively predict PM2.5.	Supported use of current-hour predictors (pollutants and meteorological) for one-hour-ahead forecasting.	RQ1 RQ2
9	Prokhorenkova et al. (2018)	CatBoost provides unbiased boosting with categorical feature handling.	Considered CatBoost as a candidate model and informed feature handling.	RQ3 RQ4
10	World Health Organization (2021)	Provided global air quality guidelines and health impacts of PM2.5.	Established the public health context and significance of PM2.5 prediction.	RQ1



Our Research Questions

RQ1

Which current-hour pollutant and meteorological variables are most strongly associated with PM2.5 concentration one hour ahead?

RQ2

How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?

RQ3

Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?

RQ4

Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?



Key Takeaways

- Prior studies informed the selection of features, models, validation methods, and interpretability approach.
- The matrix makes the RQ linkage explicit.
- This study builds on established research and addresses existing gaps by comparing eight models, using a large multi-station dataset, and providing interpretability through Gain and SHAP.



Outcome

The literature directly influenced the research questions, model choices, feature selection, and interpretability approach, ensuring this study is grounded in and extends prior research.

Research Questions

01



RQ1 – PREDICTORS

Which current-hour pollutant and meteorological variables are most strongly associated with PM2.5 concentration one hour ahead?

H₀₁: There is no significant association between current-hour pollutant and meteorological variables and one-hour-ahead PM2.5 concentration.

H₁₁: At least one current-hour pollutant or meteorological variable has a significant association with one-hour-ahead PM2.5 concentration.

03



RQ3 – MODEL PERFORMANCE

Which machine learning regression model provides the best predictive performance for one-hour-ahead PM2.5 forecasting?

H₀₃: There is no meaningful difference in predictive performance among the evaluated machine learning regression models.

H₁₃: At least one evaluated machine learning regression model demonstrates better predictive performance than the others.

02



RQ2 – FORECASTING ACCURACY

How accurately can machine learning regression models forecast PM2.5 concentration one hour ahead using current-hour pollutant, meteorological, temporal, wind-direction, and monitoring-station information?

H₀₂: Machine learning regression models do not demonstrate satisfactory predictive performance for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.

H₁₂: Machine learning regression models demonstrate satisfactory predictive performance for forecasting PM2.5 concentration one hour ahead using the selected predictor variables.

04



RQ4 – INTERPRETABILITY

Which predictor variables contribute most strongly to the forecasts of the selected machine learning model?

H₀₄: No predictor variable makes a meaningful contribution to the forecasts produced by the selected machine learning model.

H₁₄: At least one predictor variable makes a meaningful contribution to the forecasts produced by the selected machine learning model.



Evaluation Framework

Hypotheses were formulated using the conventional 5% significance framework ($\alpha = 0.05$).



Performance Metrics

MAE | RMSE | R^2



Chronological Test

Time-series train-test



Expanding-Window CV

5-fold cross-validation



Interpretability

Gain-based feature importance
+ SHAP

EDA Snapshot



412,007

observations after preprocessing



79.80 $\mu\text{g}/\text{m}^3$

mean PM2.5 concentration



$r = 0.969$

current-hour $\rightarrow$ next-hour PM2.5



DATA SOURCE

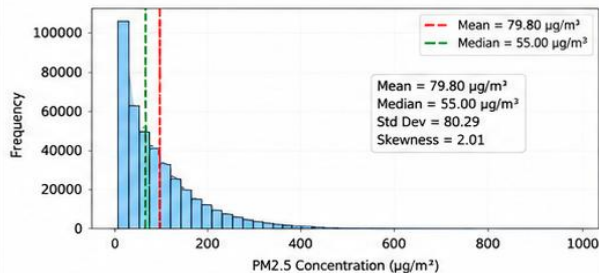
Beijing Multi-Site Air Quality Dataset (UCI) | 12 stations | Hourly | Mar 2013 – Feb 2017



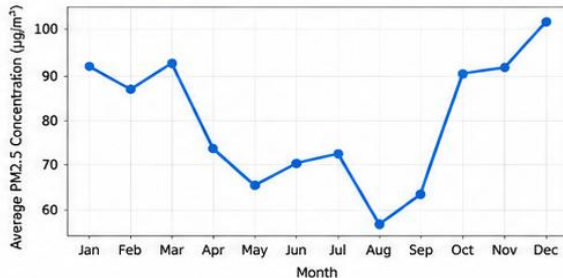
UNIT OF ANALYSIS

One station-hour observation

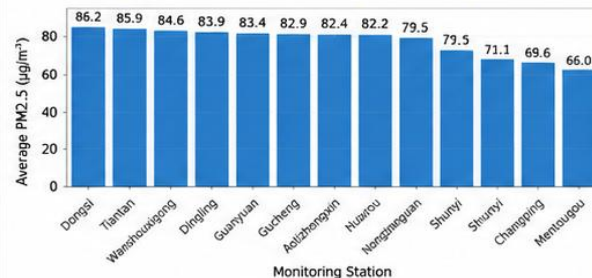
Distribution of PM2.5 Concentration



Monthly Average PM2.5 Concentration



Average PM2.5 Concentration by Monitoring Station



INSIGHT

PM2.5 is right-skewed, and its distribution indicates high-pollution episodes.



INSIGHT

Monthly averages vary substantially over time.



INSIGHT

Average PM2.5 differs across monitoring stations.



EDA $\rightarrow$ MODELLING DECISION

Strong current-hour $\rightarrow$ next-hour persistence supports the use of current-hour pollutant information, temporal, meteorological, wind-direction, and station variables for one-hour-ahead forecasting.



KEY TAKEAWAY

PM2.5 is right-skewed, varies by month and station, and current-hour PM2.5 is highly informative for one-hour-ahead forecasting.

Architecture Diagram / Workflow

End-to-end workflow for one-hour-ahead PM2.5 prediction

Objective:

Use historical pollutant, meteorological, temporal and station data to forecast next-hour PM2.5 concentration.

1



Data Collection

- Beijing multi-site dataset (12 stations, hourly)
- Mar 2013 – Feb 2017
- ~ 412,007 observations after preprocessing

2



Data Quality Checks

- Handle missing values
- Remove duplicates
- Validate timestamps
- Check outliers and inconsistencies

3



Target Variable (t+1)

- Create next-hour PM2.5 as the target variable
- Retain only exact 1-hour gaps
- Ensure no data leakage

4



Preprocessing & Feature Engineering

- Forward-fill missing values
- Create time-based features (year, month, day, hour)
- Encode categorical variables (station, wind direction)
- Use current-hour pollutant, meteorological and temporal features (X)

5



Chronological 80:20 Split

- 80% training (earlier period)
- 20% testing (later period)
- Time-based split to preserve temporal order and avoid data leakage

6



Model Comparison

8 supervised regression models using consistent model settings

- | | |
|---------------------|---------------------|
| • Linear Regression | • Random Forest |
| • Ridge Regression | • Gradient Boosting |
| • Lasso Regression | • XGBoost |
| • Decision Tree | • LightGBM |

7



Model Selection

LightGBM selected based on best performance on the unseen chronological test set (MAE, RMSE, R^2).

8



Robustness Check & Interpretability

- Five-fold expanding-window time-series cross-validation (robustness)
- Gain-based feature importance
- SHAP analysis (global and local interpretation)

9



Proposed One-Hour-Ahead Forecasting and Decision-Support Framework

Integrate the trained LightGBM model into a real-time pipeline for air-quality monitoring, early warning and environmental decision-making.

Model building

Eight supervised regression models were developed and compared to predict next-hour PM2.5.

Models evaluated



Linear
Regression



Decision
Tree



Random
Forest



Gradient
Boosting



AdaBoost



XGBoost



LightGBM



CatBoost

Why these models?

1

Linear Regression

Baseline model for comparison.

2

Tree-based models (Decision Tree, Random Forest)

Capture non-linear relationships and interactions.

3

Boosting models (Gradient Boosting, AdaBoost, XGBoost, LightGBM, CatBoost)

Often provide strong predictive performance on structured data.

4

LightGBM

Efficient gradient-boosting approach, well-suited for large tabular datasets with many features.

Model selection and evaluation

- Trained and evaluated all 8 models using a **chronological 80:20 train/test split**.
- Compared performance using **MAE, RMSE, and R^2** .
- Used consistent model settings** for a fair comparison (no extensive hyperparameter tuning).
- LightGBM achieved the **best performance on the unseen chronological test set** and was selected as the final model.
- Further validated using **5-fold expanding-window time-series cross-validation** to confirm temporal robustness.



Selected model: LightGBM

Best performance on the unseen chronological test set.

MAE

9.67

$\mu\text{g}/\text{m}^3$

RMSE

18.26

$\mu\text{g}/\text{m}^3$

R^2

0.9522

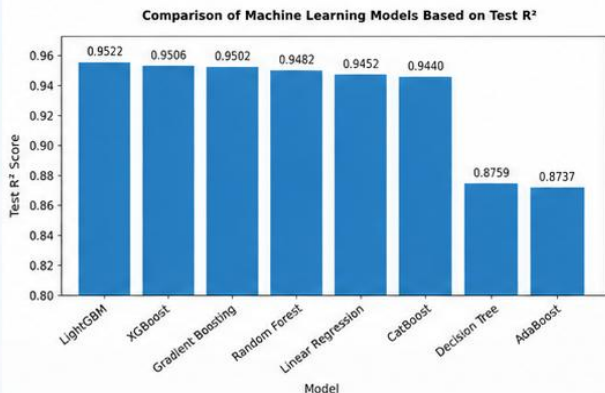


Demonstrates high predictive performance with strong temporal robustness.

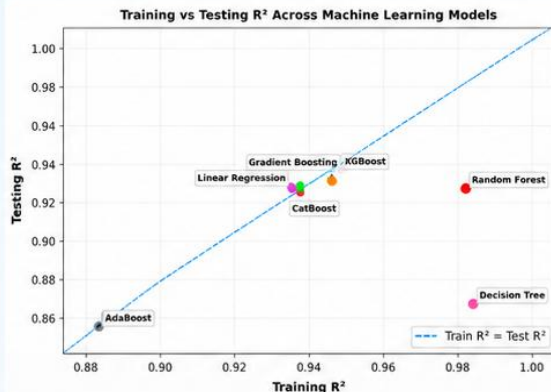
Results

Model comparison, selected-model performance, validation, and interpretability.

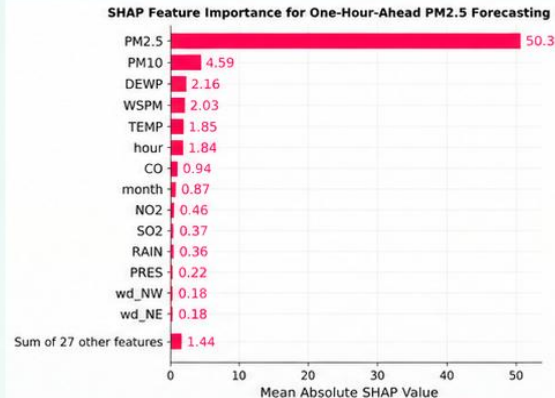
Test-set model comparison



Training vs testing R^2



SHAP feature contributions



Key findings

- 1 LightGBM ranked first on the chronological test set. [RQ3](#)
- 2 Test performance: MAE 9.67 $\mu\text{g}/\text{m}^3$ | RMSE 18.26 $\mu\text{g}/\text{m}^3$ | R^2 0.9522. [RQ2](#)
- 3 Expanding-window CV: mean MAE 11.98 | RMSE 22.43 | R^2 0.9166. [RQ2](#)
- 4 Training R^2 0.9531 vs testing R^2 0.9522 indicates no major overfitting. [RQ2](#)
- 5 PM2.5 was the dominant predictor: 98.17% of gain importance; mean |SHAP| $\approx$ 50.3. [RQ1 & RQ4](#)

Selected Model: LightGBM

Best unseen-test performance

MAE
9.67
 $\mu\text{g}/\text{m}^3$

RMSE
18.26
 $\mu\text{g}/\text{m}^3$

R^2
0.9522

Overall takeaway:

Accurate one-hour-ahead forecasting with time-aware validation and interpretable drivers.

Implementation and User Benefit

From model insights to real-world impact

Key Benefit	What it means	Real-world example / Use case
 Additional Information Source	One-hour-ahead PM2.5 forecasts can complement existing monitoring systems and provide more timely insights into air quality trends.	 - Integrate with city air-quality dashboards - Continuous monitoring for smart city initiatives
 Support for Environmental Agencies	Helps identify periods of potentially deteriorating PM2.5 conditions, enabling proactive planning and resource allocation.	 - Plan targeted inspections and mitigation actions - Allocate resources (e.g., traffic control, industrial emission checks) during high-risk periods
 Public Health Benefit	Short-term predictions can support timely air-quality advisories, helping vulnerable groups (children, elderly, respiratory patients) reduce exposure risks.	 - Issue early warnings via mobile apps, SMS or public websites - Support schools, hospitals and community outreach during high pollution episodes
 Interpretability and Trust	Gain-based importance and SHAP provide interpretable information about the key variables contributing to predictions, increasing transparency.	 - Understand key drivers (e.g., PM2.5, NO₂, temperature) - Build trust with policymakers and the public - Inform evidence-based environmental policies
 Path to Deployment	A deployment framework is proposed (e.g., real-time API or dashboard); however, independent validation and operational integration are needed.	 - Pilot deployment with historical + real-time data - Integrate into government or third-party air-quality platforms (e.g., CPCB, AQI apps) - Scalable to other pollutants and cities



Overall Real-World Impact

Enables data-driven decision-making for cleaner air, healthier communities, and more resilient, sustainable cities.



Cleaner
Environment



Healthier
Communities



More Resilient
Cities

Conclusion

Key findings, real-world significance, limitations, and future work

1

Model performance

LightGBM was the strongest of the eight evaluated models.

Test $R^2 = 0.9522$ | MAE = $9.67 \mu\text{g}/\text{m}^3$ | RMSE = $18.26 \mu\text{g}/\text{m}^3$

2

Robustness

Expanding-window validation supported temporal robustness.

Mean CV $R^2 = 0.9166$

3

Key predictive insight

Current-hour PM_{2.5} was the dominant predictor; PM₁₀, meteorological and temporal variables added information.

4

Real-world value

One-hour-ahead forecasts could complement monitoring, support earlier identification of deteriorating conditions, and inform timely air-quality advisories.

!

Limitations

Beijing-only data, one-hour horizon, limited external predictors, and no deep-learning comparison.



Future work

Multi-city validation, longer forecasting horizons, more contextual data, and LSTM / TCN / Transformer comparisons.



REAL-WORLD TAKEAWAY

Earlier information can support better air-quality decisions and healthier communities.

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